GW170817 confines the vestigial-second-sound graviton identification to one part in 10^{15} of its natural range

John G. Roehm[1]

[1] *NetRxn Foundation*

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Several emergent-gravity programs identify the spin-2 graviton with the second-sound mode of a fermionic vestigial fluid, predicting a gravitational-wave propagation speed $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ where $\chi_{\rm vest}$ is the metric-channel susceptibility of the broken-symmetry phase. We formalize this identification and confront it with the multi-messenger GW170817 [11] bound $-3 \times 10^{-15} \leq \Delta c/c \leq +7 \times 10^{-16}$ (conservatively two-sided: $|\Delta c/c| \leq 3 \times 10^{-15}$). The result is sharp: over the natural susceptibility range $\chi_{\rm vest} \in [0.1, 10]$ (a project-adopted order-unity naturalness window about the random-phase-approximation bubble-integral scale) the deviation $\Delta c/c = \sqrt{\chi_{\rm vest}} - 1$ sweeps $[-0.68, +2.16]$, vanishing only at $\chi_{\rm vest} = 1$ and exceeding the GW170817 cap by $\sim 7 \times 10^{14}$ at the endpoints. The GW170817-compatible window $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$ with $\tau = 3 \times 10^{-15}$ has measure $4\tau \approx 1.2 \times 10^{-14}$, vanishingly small under any natural-range prior. The result is a Lean 4 formal proof (`GravitationalWaves.lean`, 32 theorems, zero `sorry`, zero new axioms): a biconditional locating the compatible window plus two endpoint falsifier theorems witnessed at $\chi_{\rm vest} \in \{0.1, 10\}$, and the Phase 5y H1 underdetermination caveat. The compatible window is not disjoint from the natural range but strictly nested inside it, meeting it at $\chi_{\rm vest} = 1$; the datum *confines* $\chi_{\rm vest}$ to that window, while no speed-of-light rescaling admits either endpoint. The identification therefore survives only on a sub-interval narrower than the natural range by more than 14 orders of magnitude. The constraint is local to the leading- order kinematic identification; we identify three avenues by which emergent-gravity programs may yet recover GW170817 compatibility (symmetry-driven $\chi_{\rm vest} \to 1$ fixed point; distinct metric-channel identification decoupled from second sound; frequency-dependent $\chi_{\rm vest}$ at higher derivative order).

INTRODUCTION

Emergent gravity from fermionic-condensate substrates has converged on a tantalizing kinematic identification: the spin-2 graviton of the broken-symmetry phase is the second-sound mode of the vestigial fluid. Volovik's 2024 vestigial-fluid review [1] proposes this directly; Volovik [2] earlier made the spin counting explicit (six Goldstone modes absorbed by the spin connection in the Akama–Diakonov–Wetterich formulation [5–9]; two propagating helicities remain, identified with $h_+, h_\times$). The hydrodynamic claim is that the propagation equation of these two modes is the second-sound wave equation.

Whose claim is Eq. (1). The *identification* is Volovik's; the susceptibility-dependent speed is ours. He advances it in [3, 4] — not in the vestigial-gravity note [1], which concerns the two-step transition — and on his own de Sitter two-fluid background the Landau formula yields $s_2 = c$ *exactly*, independent of H: a structural consequence of the stiff-matter equation of state $w_n = 1$, not a tunable parameter, and hence GW170817-compatible at leading order. He leaves the spin assignment open, describing a scalar mode; the spin-2 assignment above comes from the ADW Goldstone counting and is likewise ours. Transplanting the identification onto the ADW substrate, whose metric-channel susceptibility we compute independently and which does *not* fix $\chi_{\rm vest}$, is what produces a $\chi_{\rm vest}$-dependent speed:

$$c^2_{\rm GW}(\chi_{\rm vest}) = c^2 \cdot \chi_{\rm vest}, \qquad c_{\rm GW}(\chi_{\rm vest}) = c\sqrt{\chi_{\rm vest}}, \quad (1)$$

where $\chi_{\rm vest}$ is the metric-channel susceptibility of the vestigial phase. It is a *free parameter* here, and constraining it is what this Letter does; the random-phase-approximation form our substrate models it with is set out in `VestigialSusceptibility.lean`, but no value of $\chi_{\rm vest}$ is derived there or anywhere else in the project. The underlying lattice-fermion framework is the one whose unitarity Vergeles establishes [10]. The identification is appealing in its brevity. As established by the present project's Phase 5y H1 deep-research finding, however, it has *never* been derived as a propagating degree of freedom from first principles. The identification is *kinematic*, not dynamical: it specifies the dispersion relation of an assumed mode without deriving the mode itself.

The 2017 binary-neutron-star merger GW170817 [11] placed an extraordinarily tight bound on $c_{\rm GW}$: comparing the gravitational-wave arrival time at LIGO/Virgo against the gamma-ray-burst signal at Fermi/INTEGRAL, the multi-messenger collaboration constrained $-3 \times 10^{-15} \leq \Delta c/c \leq +7 \times 10^{-16}$. Conservatively two-sided, $|\Delta c/c| \leq \tau$ with $\tau = 3 \times 10^{-15}$.

This Letter formalizes Eq. (1), places it head-to-head against GW170817, and reports the result. The conclusion is sharp: the leading-order Volovik identification survives GW170817 only on a sub-interval of the natural range $\chi_{\rm vest} \in [0.1, 10]$ narrower than that range by more than 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2,(1+\tau)^2]$ has measure $\sim 4\tau$ in $\chi_{\rm vest}$, vanishingly small under any natural-range prior, and sits strictly interior to $[0.1, 10]$ about the anchor $\chi_{\rm vest} = 1$. The result is a fine-tuning constraint on one specific kinematic identification, not a refutation of emergent gravity nor of the ADW program; we discuss three Phase 6 recovery paths in Sec. .

Relation to the 2017 constraint sweep. Using GW170817 to eliminate a gravity model is not new. Within months of the detection, four back-to-back Letters established what the bound kills: Baker *et al.* [12] covered general scalar-tensor and vector-tensor theories and bounded the graviton mass in bimetric gravity; Creminelli and Vernizzi [13] showed that avoiding tuning requires $c_{\rm GW}$ to be unaffected on *nearby* cosmological solutions, collapsing quartic and quintic beyond-Horndeski theories to a single function; Sakstein and Jain [14] combined the bound with cluster and M87 strong-equivalence-principle data to render the quartic galileon cosmologically irrelevant; and Ezquiaga and Zumalacárregui [15] extended the conclusion to Einstein-aether, Hořava, Generalized Proca and TeVeS.

The present result is not a further entry in that list, and the difference is structural rather than one of degree. Every theory the 2017 sweep reached has a Lagrangian from which a varying $c_{\rm GW}$ is *derived*; the constraint therefore lands on operator coefficients, and each Letter reports which coefficients survive. The vestigial-second-sound identification has no such Lagrangian. As Sec. notes, $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ is asserted kinematically: the mode is assumed rather than derived, so there are no operator coefficients for the bound to constrain and nothing to tune. What the bound can do instead is fix $\chi_{\rm vest}$ itself, and the content of this Letter is that it fixes $\chi_{\rm vest}$ to a part in 10^{15} of the order-unity window the identification's own microscopic origin would predict — a value that window contains, but does not favour. The 2017 sweep constrained *theories*; this constrains an *identification*, which is why it is stated as a ratio of two interval widths rather than as a surviving region of parameter space.

A corollary worth stating plainly: because the argument never uses a Lagrangian, it is insensitive to the operator-level escape routes those Letters catalogue. Conformal rescaling, coefficient compensation and restriction to Horndeski's simplest terms all act on terms that the kinematic identification does not have. The three recovery paths in Sec. are correspondingly different in kind, and each requires replacing the identification rather than repairing it.

FORMAL SETUP AND RESULT

Vestigial-susceptibility natural range. Our substrate models the vestigial-phase metric-channel susceptibility by the RPA *resummation* $\chi_{\rm RPA} = \chi_0/(1-\gamma_\star\chi_0)$ in the standard Hertz–Millis convention, where χ_0 is the bubble integral — the two are distinct, and $\chi_{\rm RPA}$ is not itself a bubble integral. `VestigialSusceptibility.lean` formalizes that closed form and its stability properties; it evaluates neither χ_0 nor $\gamma_\star$, so it fixes no numerical $\chi_{\rm vest}$, which is why $\chi_{\rm vest}$ enters Eq. (1) as a free parameter. The natural range $\chi_{\rm vest} \in [0.1, 10]$ (in units of inverse $\Lambda^2_{\rm UV}$) is a *project-adopted* naturalness window: the one-loop bubble normalization is $\mathcal{O}(\Lambda^2_{\rm UV}/16\pi^2)$ by dimensional analysis, and we take the ± 1-decade window about the resulting unit-normalized central value. We emphasize the provenance: the window is a modeling choice of this project, not a published derivation: the Vergeles lattice analysis [10] establishes unitarity of the underlying Einstein–Cartan–Palatini lattice theory but does not itself compute $\chi_{\rm vest}$ or assign it a range. The window matches the linearized-EFE wave's ADW microscopic-coefficient $\alpha_{\rm ADW} \in [0.1, 10]$ natural range [16], which we treat as the project-internal anchor for the order-of-magnitude prior.

The leading-order $c_{\rm GW}$. Equation (1) ships in `GravitationalWaves.lean` as `c_GW(chi, c) := c * Real.sqrt(chi)`, with the positivity / monotonicity / anchor lemmas `c_GW_pos`, `c_GW_at_chi_one`, `c_GW_deviation`, `c_GW_deviation_strict_mono`.

The GW170817 correctness-push. Define $\Delta c/c(\chi_{\rm vest}) = \sqrt{\chi_{\rm vest}} - 1$ and the predicate `LigoSatisfied`$(\delta) \equiv |\delta| \le \tau$. The correctness-push biconditional is

Theorem 1 (`c_GW_match_iff_chi_close_to_one`)**.** *For* $\chi_{\rm vest} \ge 0$, `LigoSatisfied`$(\Delta c/c(\chi_{\rm vest})) \iff \chi_{\rm vest} \in [(1-\tau)^2,(1+\tau)^2]$.

The proof is a $\sqrt{\cdot}$/squaring round-trip via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, with `nlinarith` supplied the hint $\sqrt{\chi_{\rm vest}}^2 = \chi_{\rm vest}$. The compatible $\chi_{\rm vest}$ window has width $\sim 4\tau = 1.2\times 10^{-14}$.

Natural-range fine-tuning. The natural range $[0.1, 10]$ has width 9.9 (`natural_range_width_eq`); the GW170817-compatible window has width exactly $4\tau = 1.2\times10^{-14}$ (`ligo_window_width_eq`) and is nested strictly inside it (`ligo_window_subset_natural_range`), not disjoint from it. Scaling the window up by 10^{14} still leaves it the narrower (`ligo_window_width_lt_natural_range_width_by_1e14`): that ratio, not a disjointness, is the fine-tuning claim. Two endpoint-falsifier theorems sharpen the exclusion of the extremes.

Theorem 2 (`natural_lower_violates_ligo`)**.** *At* $\chi_{\rm vest} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, *hence* $|\Delta c/c|/\tau \approx 2.28\times10^{14}$ *and* `LigoSatisfied` *is false.*

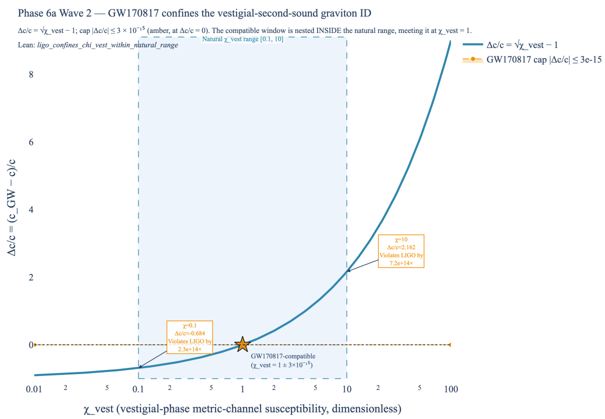

FIG. 1. **Headline result.** The $\Delta c/c = \sqrt{\chi_{\rm vest}} - 1$ deviation across the natural $\chi_{\rm vest} \in [0.1, 10]$ range (pale blue band, dashed blue border) versus the GW170817 [11] multi-messenger cap $|\Delta c/c| \leq 3 \times 10^{-15}$ (amber, drawn at $\Delta c/c = 0$ — the cap's own width is far below one pixel at this scale). Over the natural range $\Delta c/c$ sweeps $[-0.68, +2.16]$, exceeding the cap by $\sim 10^{14}\times$ *at the endpoints* (orange callouts) and vanishing at $\chi_{\rm vest} = 1$. The compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ is 10^{14} times narrower than the natural range and so renders as the single gold star — note that it sits *inside* the band, not beside it: the datum confines $\chi_{\rm vest}$, it does not exclude the range. Lean witnesses: `ligo_confines_chi_vest_within_natural_range`, `ligo_window_subset_natural_range`, `vestigial_natural_range_violates_ligo`.

Theorem 3 (`natural_upper_violates_ligo`). *At $\chi_{\rm vest} = 10$, $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$, hence $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$ and* `LigoSatisfied` *is false.*

The proofs use $\sqrt{1/10} \leq \sqrt{1/4} = 1/2$ and $\sqrt{10} \geq \sqrt{4} = 2$, both via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, closed by `linarith`. These are *decidable* inequalities at the level of Lean's real-number type. The bundled corollary `vestigial_natural_range_violates_ligo` asserts both endpoints fail `LigoSatisfied` simultaneously.

The identification hypothesis itself is refuted pointwise rather than only at the observable: `H_VestigialModeIsGraviton_fails_at_natural_lower` and `H_VestigialModeIsGraviton_fails_at_natural_upper` refute it at the two endpoints, and `H_VestigialModeIsGraviton_fails_at_zero` and `H_VestigialModeIsGraviton_fails_at_quarter` refute it at $\chi_{\rm vest} \in \{0, 1/4\}$, which are outside the natural range and establish that the failure is not an endpoint artefact. The one value that survives is the singleton: `ligo_satisfied_at_chi_one` proves `LigoSatisfied` holds at $\chi_{\rm vest} = 1$, and `c_GW_deviation_zero_iff_chi_one` makes that the exact biconditional.

Tracked-hypothesis bundle. Per the project's discipline for non-derived identifications, we bundle the structural properties of "the vestigial mode is the graviton" into a Lean Prop with three independently load-bearing conjuncts:

$$\texttt{H\_VestigialModeIsGraviton}(\chi_{\rm vest}) \equiv \chi_{\rm vest} > 0 \wedge \texttt{LigoSatisfied}(\Delta$$

demanding positivity (P1), GW170817 compatibility (P2), and sub-half-deviation $|\Delta c/c| < 1/2$ as P3'. P3' is genuinely independent of P1 (e.g. $\chi_{\rm vest} = 1/10$ satisfies P1 with $|\Delta c/c| \approx 0.684 \not< 1/2$). The bundle is discharged at $\chi_{\rm vest} = 1$ (`H_VestigialModeIsGraviton_at_one`) and *four* explicit falsifier theorems prove non-vacuity: `*_fails_at_natural_lower` (P2), `*_fails_at_natural_upper` (P2), `*_fails_at_zero` (P1), and `*_fails_at_quarter` (P3'; at $\chi_{\rm vest} = 1/4$ the deviation is exactly $\Delta c/c = -1/2$, saturating the bound and isolating P3' independence). The bundle's support over the natural-range prior has measure $4\tau \approx 1.2 \times 10^{-14}$ — vanishingly small, but non-empty, and containing $\chi_{\rm vest} = 1$ — and that is the load-bearing falsifiability claim. (The original Wave-2 P3 = $\forall c{>}0,\, c_{\rm GW}(\chi_{\rm vest}, c) > 0$ was removed in the 2026-04-25 post-Wave-2 audit as redundant given P1; P3' replaces it with a load-bearing quantitative constraint per the project's preemptive-strengthening discipline.)

Phase 5y H1 caveat as a Lean endpoint theorem. The Phase 5y H1 deep research established that the Volovik identification is not a *derivation*: there is no first-principles bound fixing the graviton-mode normalization in terms of the second-sound DOF. We formalize this caveat as the endpoint theorem `natural_endpoints_violate_ligo_window`: for $\chi_{\rm vest}$ at either natural-range endpoint $\{0.1, 10\}$, no c-rescaling gives a GW170817-compatible $\Delta c/c$. The frame-independent form `natural_endpoints_violate_ligo_for_every_c` quantifies over all $c > 0$ explicitly, resting on `c_GW_relative_deviation_eq`. Both concern the two endpoints only; neither excludes interior values, and $\chi_{\rm vest} = 1$ satisfies the cap (`ligo_satisfied_at_chi_one`).

DISCUSSION

What this rules out. GW170817 does not exclude the leading-order vestigial-second-sound identification outright: it survives at $\chi_{\rm vest} = 1$. What it rules out is the identification as a *generic* consequence of the natural range. Eq. (1) requires $\chi_{\rm vest} = 1 \pm 3 \times 10^{-15}$, a fine-tuning so extreme it cannot follow from a generic UV theory — so any surviving version of the identification owes an account of why $\chi_{\rm vest}$ sits exactly at unity.

What this does not rule out. Three Phase 6 paths remain open:

1. **Symmetry-driven $\chi_{\rm vest} \to 1$ fixed point.** If a deep-IR symmetry (e.g. Lorentz invariance of the

emergent metric sector) makes $\chi_{\rm vest} = 1$ an exact fixed point of the RG flow, the fine-tuning is structural rather than tuned. The present evidence most favours this path: Volovik's own background returns $s_2 = c$ exactly, i.e. $\chi_{\rm vest} = 1$, structurally. The value GW170817 selects is thus the value his construction already produces — the datum corroborates his result and indicts our substrate's failure to reproduce it, not the identification. Phase 6 should examine whether such a fixed point is realized in the ADW broken phase.

2. **Distinct metric-channel identification.** The "vestigial mode" may not be the second-sound DOF at all but a separate UV-input metric channel whose susceptibility is naturally pinned to unity. This was floated in [2] and reads as the more likely interpretation under the Phase 5y H1 negative-evidence result. Phase 6e (full nonlinear emergent action) is the natural venue for the disambiguation.

3. **Frequency-dependent $\chi_{\rm vest}$.** Eq. (1) is the leading term in a derivative expansion. If $\chi_{\rm vest}$ varies with frequency, the leading-order constraint could in principle be relaxed by a near-cancellation in the LIGO band. The SK–EFT dispersion correction is the leading frequency dependence in the present framework, and this path is *not* closed — what we can say is conditional. `vestigial_dispersion_below_ligo_at_inspiral_peak` takes $|\gamma| \leq 10^{-30}$ on $\Gamma_H/c_{\rm GW}^2$ as a *hypothesis* and proves that under it the correction at the 100 Hz inspiral peak stays inside the GW170817 tolerance. It proves nothing about whether that bound holds: 10^{-30} is a Phase 6a scaffold value, chosen so the correction would sit well below the cap, and it is neither derived nor measured. So a placeholder-sized correction would not relax the constraint; a larger one is not excluded. The general form is `dispersion_within_ligo_iff`, a biconditional exchanging the correction's tolerance for $|\gamma| \leq \mathrm{tol}/\omega$; `dispersion_correction_abs_bound`, `dispersion_correction_linear_in_gamma` and `dispersion_correction_zero_at_no_dissipation` fix its magnitude, additivity and no-dissipation limit. A derivation of Γ_H from a microscopic matching is open.

Comparison to the linearized-EFE wave. The companion linearized-EFE wave [16] ships an emergent-gravity identification *compatible* with observation at the natural-range level: the Sakharov–Adler form $G_{\rm N,emerg} = \alpha_{\rm ADW} \cdot 12\pi/(N_f \Lambda_{\rm UV}^2)$ at $\Lambda_{\rm UV} = M_{\rm Pl}$ produces $\alpha^\star_{\rm ADW} \in [0.40, 1.27]$ for SM N_f, sitting inside the natural range $[0.1, 10]$. The asymmetry between the two waves is a feature of the program: the linearized-EFE $G_{\rm N,emerg}$ depends on the Sakharov–Adler kernel computed at the linearized level, where the relevant scale is $M_{\rm Pl}$; $c_{\rm GW}$ depends on the metric-channel susceptibility, where the relevant scale is the dimensionless $\chi_{\rm vest}$ which the natural range happens to pin around 1 only by coincidence. GW170817 sees the latter, and its bound is too tight to absorb the natural-range spread.

Reproducibility. The result is reproducible in two layers. The Lean layer is `GravitationalWaves.lean` (32 theorems, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project's pinned Mathlib4 commit; the Python layer is `src/gravitational_waves/` with 49 `pytest` cases cross-checking every numeric quoted in the abstract and figure caption. The headline figure (`src/core/visualizations.py:fig_c_GW_vs_ligo_constraint`) and its underlying numerics are committed-pinned.

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